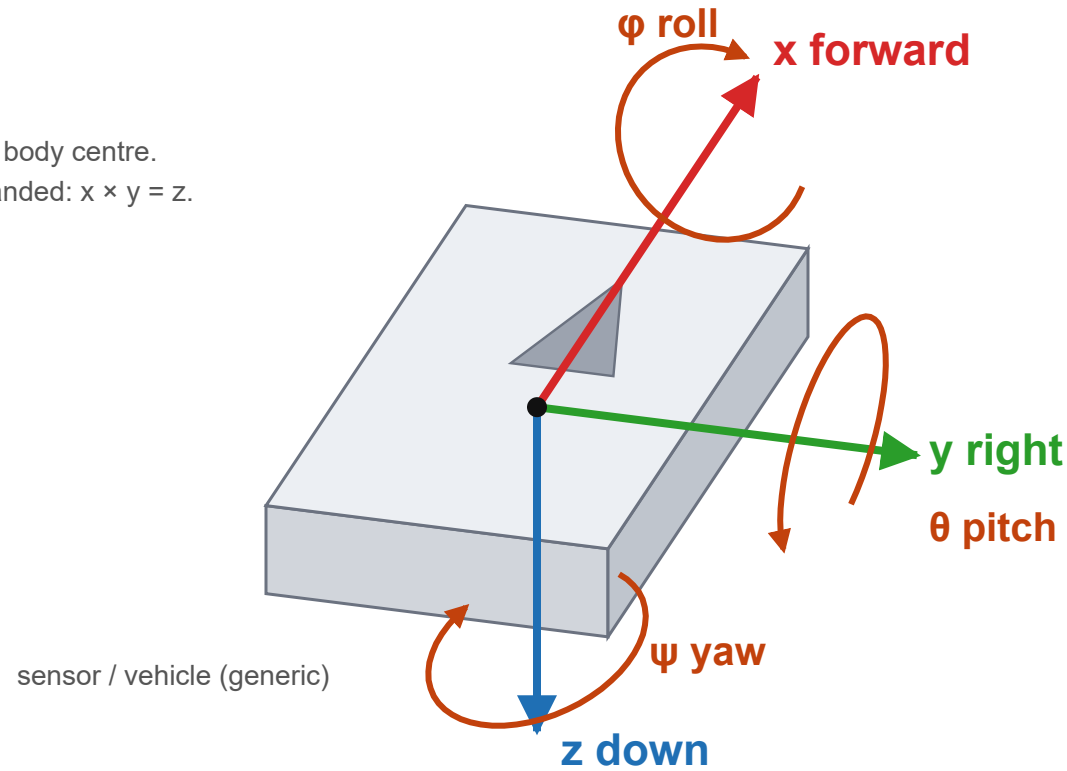
INSLIB body frame: FRD (Forward - Right - Down)

Origin = body centre.
Right-handed: x × y = z.



Rules for all INSLIB inputs

x forward (nose)
y right (right wing)
z down (towards the ground)

Positive rotation = right-hand rule
roll φ about x: right side goes down
pitch θ about y: nose goes up
yaw ψ about z: nose turns right

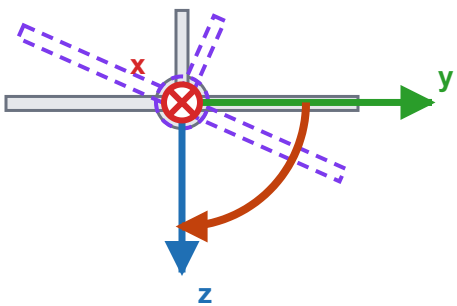
Gyroscope
Rate about the body axes in rad/s.

Accelerometer
Specific force in m/s², not g. Level and at rest it reads (0, 0, **-9.81**), not +9.81.

Sensor mounted differently?
Rotate into FRD first (e.g. FLU: y→-y, z→-z).

Roll φ about x

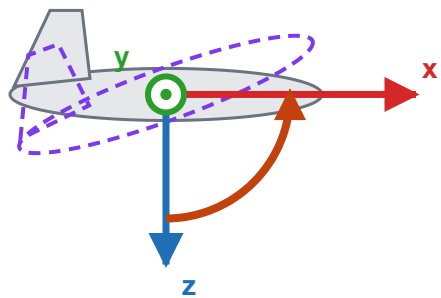
view from behind, x points into the page



right wing down = positive

Pitch θ about y

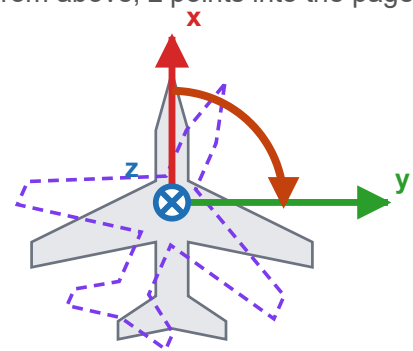
view from the right, y points at you



nose up = positive

Yaw ψ about z

view from above, z points into the page



nose to the right = positive